LAB 20

Capstone: One Command Runs the Pipeline End to End, with an Agentic Layer and Human Approval

Item	Detail
Topic	Topic 4: Building an End-to-End Algorithmic Trading Pipeline: Backtest, Forward Test and Performance Analysis
Pipeline stage	End-to-end pipeline
TSC reference	A7, A10, K1
Duration	20 min
Simulator page	http://localhost:8000/#/agents
Mode	No coding. Runs ONLINE with your Alpaca paper keys (Integrations page), or OFFLINE on the demo data and the built-in paper broker. Paper trading only.

Lion City Capital and every scenario in these materials are fictitious and for training only. All trading in this course is paper (simulated) trading. Nothing in these materials is investment advice or a recommendation to trade any security.

Scenario

Lion City Capital's head of trading wants the whole pipeline - data, research, signals, execution, automation, diagnostics and reporting - to run with one command, stop sending orders whenever a gate fails, and finish with an optional multi-agent layer (inspired by the TradingAgents framework, arXiv 2412.20138) whose proposals pass hard risk limits and a human's approval before any paper order.

Objective

Run the whole pipeline end to end, get a human-approved proposal from the multi-agent desk, switch on the always-on agent, and create, backtest and forward-test your own strategy (from a template or written for you by AI).

Before You Start

- Make sure the Trading Simulator is running and open http://localhost:8000 > Course Labs > Lab 20. It is installed once - see the guide 'Download, Install and Start the Trading Simulator Software' in the Learner Guide.
- Record every result you are asked to check - the same evidence is asked for in the Practical Performance assessment.

Lab Steps - In the Trading Simulator

No coding is needed. Make sure the Trading Simulator is running (installed once - see the guide 'Download, Install and Start the Trading Simulator Software') and open http://localhost:8000. In the sidebar open Course Labs > Lab 20: the app shows the same steps with a Go button that opens each page pre-filled. Every step works offline on the bundled demo data and the built-in paper broker.

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Step 1. Run the multi-agent desk

AI Agents > Run multi-agent analysis on NVDA, AAPL, SPY. Read each agent's card: three analysts, the two-round bull/bear debate, the trader, the risk manager, the hard-limit gate (code) and the portfolio manager.

Open: http://localhost:8000/#/agents (AI Agents > desk)

Check: Eight agent entries per symbol plus the code gate; the brain badge shows which provider or rules answered.

Step 2. Human approval

Approve (or reject) the proposals. Approved buys become paper bracket orders in the regular session or extended-hours limit orders overnight.

Open: http://localhost:8000/#/agents

Check: Run status 'executed' and the orders visible in Paper Trading (source agent).

Step 3. Always-on agent

AI Agents > Always-on agent: enable it with a 60-minute wake-up (the worker must run), or press 'Wake it now'. Check Memory.

Open: http://localhost:8000/#/agents

Check: A new always_on run and memory notes appear.

Step 4. Create your own strategy

Strategy Builder: start from a template and change only its parameters - or describe your idea in the Lab 20 prompt of the Prompt Library and paste the code the AI writes. Validate (look-ahead check), Save.

Open: http://localhost:8000/#/builder?template=ema_rsi (Strategy Builder (EMA + RSI template))

Check: Your strategy appears in the Strategy Library with an auto-generated parameter form.

Step 5. Backtest, diagnose and forward-test it

Backtest it, run Diagnostics, then create a paper bot for it and start it.

Open: http://localhost:8000/#/trading?new=1

Check: Pipeline Map > Strategy lifecycle for your strategy shows the gates it passed.

Step 6. End-to-end on the Pipeline Map

Open the Pipeline Map and press 'Run a bot cycle'. Then switch to 'Strategy lifecycle'.

Open: http://localhost:8000/#/pipeline (Pipeline Map)

Check: All 14 stages have run; you can walk a colleague through each stage.

Expected in the app: A Pipeline Map with every stage green, an approved agent proposal with its decision trace, and your custom strategy trading on paper.

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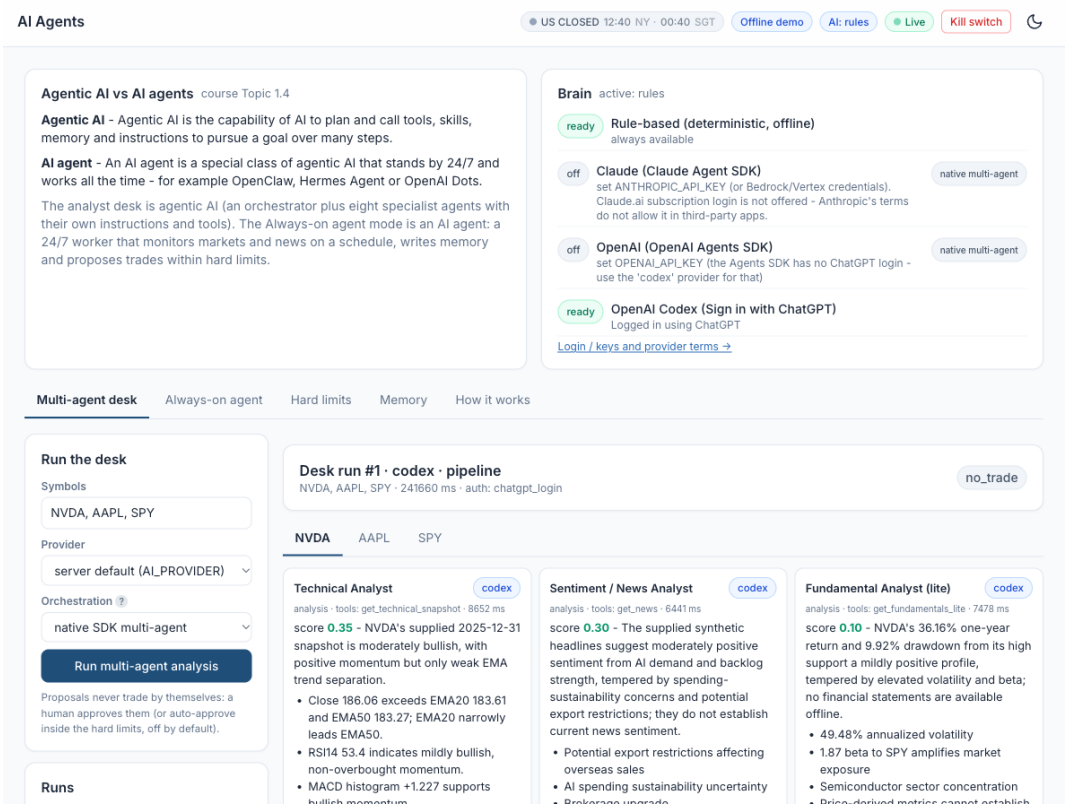


Figure: Lab 20 in the Trading Simulator.

Optional: Let AI Write the Code

This lab needs no coding. To see, run or customise the code behind it, click a link: the AI opens with the prompt filled in and writes the code for you. All prompts are also in labs_lowcode/Prompt_Library.html and .pdf.

Your own strategy for the Strategy Builder - paste it into the Strategy Builder
Write a strategy for my Trading Simulator's Strategy Builder. Rules: one Python class that subclasses Strategy (from trader.strategies.base import Strategy, enter_exit, flat_until_ready); import indicators with `from trader.core import indicators as ta` (ta.sma(series, window), ta.ema(series, span), ta.rsi(close, window), ta.macd(close, fast, slow, signal) returns columns macd/signal/hist, ta.bollinger(close, window, k) returns columns lower/middle/upper/pct_b, ta.atr(df, window), ta.donchian(df, window), ta.realized_vol(close, window)); give it a name and a params dict such as {"window": {"default": 20, "min": 2, "max": 200, "help": "Lookback in bars"}}; implement generate_signals(self, data) where data is an OHLCV DataFrame, and return a Series of 1 (long), 0 (flat) or -1 (short) aligned to data.index. Use only data up to each bar - no center=True and no shift(-1) - because the engine trades on the next bar. Reply with the code only, then 3 bullet points explaining the rule. My idea: buy SPY when RSI(14) falls below 30 while the close is above its 200-day SMA, and sell when RSI(14) rises above 55. (Replace this idea with your own.)

Open in Claude | Open in ChatGPT

Debrief

Agentic AI (K1) chains specialised agents that reason, debate and act, which makes it powerful and risky. Wiring it into an end-to-end pipeline whose gates, deterministic risk limits, kill switch and human approval it cannot override (A7), and evaluating every stage's evidence critically (A10), is how a desk adopts AI and automation responsibly.

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